TITAN research-software demonstration

24 August 2026

Contents

Research-software output: internally derived TCGA estimates

COAD - TCGA-AA-A01F

UNVALIDATED RESEARCH OUTPUT <U+2014> BINARY TCGA OOF SCORE RANK (NOT PROBABILITY) <U+2014> NO INDEPENDENT EXTERNAL VALIDATION <U+2014> NOT FOR DIAGNOSIS, TREATMENT SELECTION OR CLINICAL RISK ESTIMATION

Models applied	Continuous	Binary	Slides pooled
19	10	9	1

This research-software demonstration applies models developed and evaluated only in TCGA. Every displayed value is an **internally derived TCGA estimate**, not external performance evidence or a diagnostic output. Binary LDA scores are uncalibrated. Every displayed **TCGA OOF score rank (not probability)** describes relative position only.

By default, TITANPred excludes binary models with fewer than 50 participants in either development class. Such models appear only after the caller explicitly sets `include_limited_evidence = TRUE`.

Prominent tissue-source-site sensitivity warning

12 model(s) in this output fell below their original prespecified screening threshold under TCGA tissue-source-site-grouped internal validation. These calls must not be highlighted without this warning. Tissue-source-site grouping is not institutional or scanner-level external validation.

Endpoint (type)	TSS-grouped metric	Grouped minus random	Warning category
APC (binary)	BA 0.543	-0.250	Near chance or worse
PI3K (binary)	BA 0.593	-0.020	Below prespecified scr
Aneuploidy score (continuous)	Q2 0.173	-0.086	Below prespecified scr
Deleted arm count (continuous)	Q2 0.147	-0.066	Below prespecified scr
MANTIS score (continuous)	Q2 0.113	-0.104	Below prespecified scr
MSIsensor score (continuous)	Q2 0.026	-0.229	Below prespecified scr
Aneuploidy Score (continuous)	Q2 0.174	-0.096	Below prespecified scr
Lymphocyte Infiltration Signature Score (continuous)	Q2 0.048	-0.238	Below prespecified scr
Nonsilent Mutation Rate (continuous)	Q2 0.088	-0.127	Below prespecified scr
SNV Neoantigens (continuous)	Q2 -0.013	-0.252	Near chance or worse
Silent Mutation Rate (continuous)	Q2 -0.003	-0.236	Near chance or worse
TIL Regional Fraction (continuous)	Q2 0.176	-0.278	Below prespecified scr

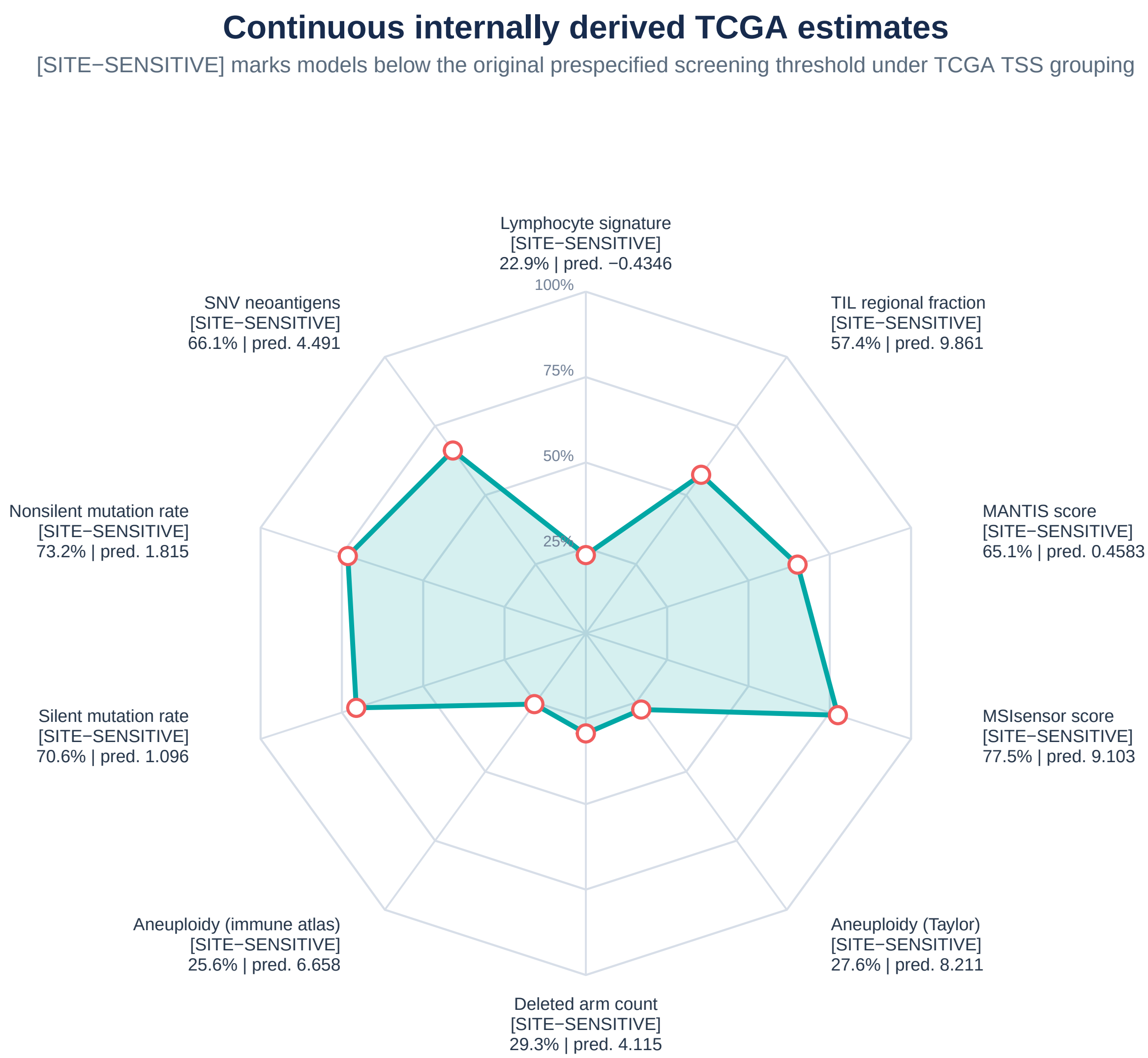
Case context (not used as model input)

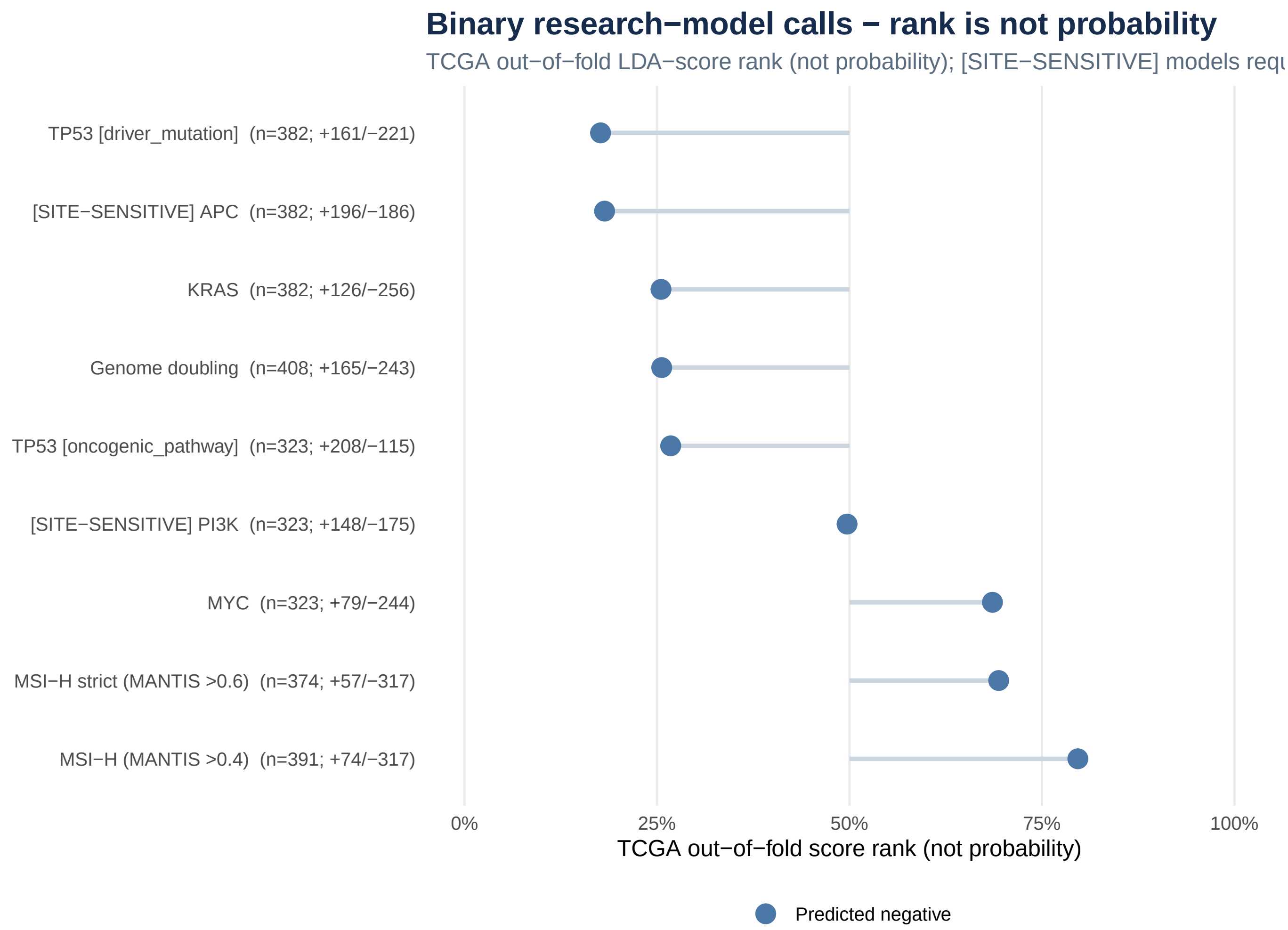
Shared selection features. Both cases were men with sigmoid-colon, moderately differentiated (G2), pT3 adenocarcinoma and tumour-free resection margins. Nodal status was not matched: example A was pN1, whereas the available slide summary for example B did not state a nodal category.

Pathology summary. The sigmoid resection contained an ulcerated grade-2 adenocarcinoma that had crossed the muscular wall into pericolic fat (pT3). Both resection ends were free of carcinoma; lymphatic invasion was recorded, and 2 of 30 regional nodes contained metastasis (pN1).

Clinical-context source. Pathology was paraphrased from TCGA-Slide-Reports.csv distributed with the TITAN study (Ding et al., Nature Medicine 2025; doi:10.1038/s41591-025-03982-3).

The pathology field is descriptive context only; it was not a predictor, training target, or validation outcome for TITANPred. Treatment and response narratives are intentionally excluded from this research-software output.





Continuous internally derived estimates

Family	Endpoint	Prediction	TCGA OOF prediction percentile (not prob
microsatellite_instability	MSIsensor score	9.103	77.5%
thorsson	Lymphocyte Infiltration Signature Score	-0.4346	22.9%
thorsson	Aneuploidy Score	6.658	25.6%
thorsson	Nonsilent Mutation Rate	1.815	73.2%
aneuploidy	Aneuploidy score	8.211	27.6%
aneuploidy	Deleted arm count	4.115	29.3%
thorsson	Silent Mutation Rate	1.096	70.6%
thorsson	SNV Neoantigens	4.491	66.1%
microsatellite_instability	MANTIS score	0.4583	65.1%
thorsson	TIL Regional Fraction	9.861	57.4%

Internal development evidence. The primary screen and five-repeat estimates use different nested-CV partitions and need not be identical.

Endpoint	Primary screen Q2	Five-repeat Q2	Five-repeat RMSE	Five-repeat Spearman
Aneuploidy score	0.260	0.294	6.879	0.570
Deleted arm count	0.213	0.196	5.742	0.565
MANTIS score	0.217	0.243	0.201	0.392
MSIsensor score	0.254	0.218	9.707	0.269
Aneuploidy Score	0.269	0.272	5.936	0.555
Lymphocyte Infiltration Signature Score	0.285	0.285	0.726	0.532
Nonsilent Mutation Rate	0.214	0.247	0.805	0.335
SNV Neoantigens	0.239	0.208	0.969	0.368
Silent Mutation Rate	0.234	0.242	0.651	0.351
TIL Regional Fraction	0.454	0.423	6.428	0.688

Binary research-model calls

Calls and development denominators.

Endpoint	Class	LDA score	TCGA OOF score rank (not probability)	Training n (+/-)
Genome doubling	0	-2.153	25.6%	408 (+165/-243)
APC	0	-2.982	18.2%	382 (+196/-186)
KRAS	0	-3.159	25.5%	382 (+126/-256)
TP53 [driver_mutation]	0	-3.171	17.7%	382 (+161/-221)
MSI-H (MANTIS >0.4)	0	-1.165	79.7%	391 (+74/-317)
MSI-H strict (MANTIS >0.6)	0	-4.352	69.4%	374 (+57/-317)
MYC	0	-1.023	68.6%	323 (+79/-244)
PI3K	0	-0.2634	49.7%	323 (+148/-175)
TP53 [oncogenic_pathway]	0	-0.05124	26.8%	323 (+208/-115)

Internal development evidence. Primary-screen BA and five-repeat estimates use different nested-CV partitions and need not be identical. PPV and NPV use the observed TCGA prevalence.

Endpoint	Primary screen BA	Five-repeat BA	Five-repeat AUROC	Five-repeat PR-AUC	PPV
Genome doubling	0.691	0.692	0.760	0.678	0.636
APC	0.793	0.794	0.848	0.814	0.760
KRAS	0.680	0.692	0.781	0.581	0.600
TP53 [driver_mutation]	0.779	0.778	0.854	0.793	0.718
MSI-H (MANTIS >0.4)	0.744	0.732	0.834	0.634	0.677
MSI-H strict (MANTIS >0.6)	0.784	0.782	0.915	0.741	0.750
MYC	0.603	0.605	0.676	0.417	0.515
PI3K	0.613	0.607	0.640	0.583	0.582
TP53 [oncogenic_pathway]	0.671	0.668	0.734	0.815	0.752

Binary development reliability

Fold class counts and component selection.

Endpoint	Minimum outer test (+/-)	Minimum inner training (+/-)	Components: median (range)
Genome doubling	+33 / -48	+105 / -155	6.0 (4-10)
APC	+39 / -37	+124 / -118	2.0 (1-4)
KRAS	+25 / -51	+80 / -163	4.0 (2-10)
TP53 [driver_mutation]	+32 / -44	+102 / -140	2.0 (2-7)
MSI-H (MANTIS >0.4)	+14 / -63	+47 / -202	6.0 (4-9)
MSI-H strict (MANTIS >0.6)	+11 / -63	+36 / -202	7.0 (3-10)
MYC	+15 / -48	+50 / -156	4.0 (2-5)
PI3K	+29 / -35	+94 / -112	4.0 (2-10)
TP53 [oncogenic_pathway]	+41 / -23	+132 / -73	3.0 (1-6)

Repeat-to-repeat held-out prediction stability.

Endpoint	Score Spearman rho	Class agreement
Genome doubling	0.850	0.857
APC	0.945	0.944
KRAS	0.825	0.858
TP53 [driver_mutation]	0.948	0.931
MSI-H (MANTIS >0.4)	0.856	0.940
MSI-H strict (MANTIS >0.6)	0.804	0.948
MYC	0.861	0.905
PI3K	0.721	0.787
TP53 [oncogenic_pathway]	0.853	0.895

Quality and provenance

- Patient-level aggregation: arithmetic mean across all supplied slide vectors.
- Maximum fraction outside endpoint-specific TCGA training ranges: 0.3%.
- TITAN input schema: 768 named features (`titan_000`–`titan_767`).
- Decomposition: seeded rSVD in fastPLS; binary classification: PLS-LDA.
- Tissue-source-site metadata: every row includes the grouped internal metric, random-minus-grouped change, analysed-site count, threshold-retention status, fold-size range and inner-site-separation audit.
- Binary reliability metadata: held-out average precision, cohort-specific PPV/NPV, outer and inner class-count minima, selected-component distribution and repeat-to-repeat prediction stability.
- Default model selection: binary models require at least 50 TCGA participants in each class; 17 smaller-class models are available only by explicit opt-in.
- Scope: TCGA tissue-source-site-grouped internal validation is not institutional or scanner-level external validation.
- Intended use: research-software demonstration only; TCGA benchmark without independent external validation.

Endpoint interpretation and target sources

For continuous endpoints, the TCGA OOF prediction percentile (not probability) locates the prediction within repeated TCGA out-of-fold predictions and is not a clinical reference interval. For binary endpoints, every TCGA OOF score rank (not probability) is the relative position of an uncalibrated LDA score. It is not confidence or a clinical-risk estimate. PR-AUC is non-interpolated average precision for positive class 1. PPV and NPV are held-out estimates at the observed endpoint prevalence in this TCGA analysis and will change when prevalence changes. Rows marked exploratory/limited evidence are excluded from default inference and require explicit opt-in.

- **Genome doubling.** Binary whole-genome-doubling call derived from the tumour copy-number profile. *Analysed scale:* PLS-LDA class and uncalibrated discriminant score *Model-target source:* Taylor et al., Cancer Cell 2018, Table S2; doi:10.1016/j.ccell.2018.03.007.
- **APC.** Binary prediction of a qualifying protein-altering PASS mutation in APC; the displayed LDA score is not a probability. *Analysed scale:* PLS-LDA class and uncalibrated discriminant score *Model-target source:* MC3 consensus calls: Ellrott et al., Cell Systems 2018, doi:10.1016/j.cels.2018.03.002; cancer-gene eligibility: Bailey et al., Cell 2018, doi:10.1016/j.cell.2018.02.060.
- **KRAS.** Binary prediction of a qualifying protein-altering PASS mutation in KRAS; the displayed LDA score is not a probability. *Analysed scale:* PLS-LDA class and uncalibrated discriminant score *Model-target source:* MC3 consensus calls: Ellrott et al., Cell Systems 2018, doi:10.1016/j.cels.2018.03.002; cancer-gene eligibility: Bailey et al., Cell 2018, doi:10.1016/j.cell.2018.02.060.
- **TP53 [driver_mutation].** Binary prediction of a qualifying protein-altering PASS mutation in TP53; the displayed LDA score is not a probability. *Analysed scale:* PLS-LDA class and uncalibrated discriminant score *Model-target source:* MC3 consensus calls: Ellrott et al., Cell Systems 2018, doi:10.1016/j.cels.2018.03.002; cancer-gene eligibility: Bailey et al., Cell 2018, doi:10.1016/j.cell.2018.02.060.
- **MSI-H (MANTIS >0.4).** Binary call indicating whether the source MANTIS score exceeds 0.4; the displayed LDA score is not a probability. *Analysed scale:* PLS-LDA class and uncalibrated discriminant score *Model-target source:* Bonneville et al., JCO Precision Oncology 2017; doi:10.1200/PO.17.00073.
- **MSI-H strict (MANTIS >0.6).** Sensitivity-analysis call: MANTIS >0.6 is positive, <0.4 is negative, and intermediate source values were excluded from training. *Analysed scale:* PLS-LDA class and uncalibrated discrim-

inant score *Model-target source:* Threshold documentation distributed with the cBioPortal TCGA PanCancer Atlas clinical sample files.

- **MYC.** Binary prediction that the MYC oncogenic pathway carries at least one qualifying genomic alteration; the displayed LDA score is not a probability. *Analysed scale:* PLS-LDA class and uncalibrated discriminant score *Model-target source:* Sanchez-Vega et al., Cell 2018, Table S4; doi:10.1016/j.cell.2018.03.035.
- **PI3K.** Binary prediction that the PI3K oncogenic pathway carries at least one qualifying genomic alteration; the displayed LDA score is not a probability. *Analysed scale:* PLS-LDA class and uncalibrated discriminant score *Model-target source:* Sanchez-Vega et al., Cell 2018, Table S4; doi:10.1016/j.cell.2018.03.035.
- **TP53 [oncogenic_pathway].** Binary prediction that the TP53 oncogenic pathway carries at least one qualifying genomic alteration; the displayed LDA score is not a probability. *Analysed scale:* PLS-LDA class and uncalibrated discriminant score *Model-target source:* Sanchez-Vega et al., Cell 2018, Table S4; doi:10.1016/j.cell.2018.03.035.
- **Aneuploidy score.** Number of chromosome arms with arm-level copy-number gain or loss; larger values indicate greater chromosomal imbalance. *Analysed scale:* source endpoint units *Model-target source:* Taylor et al., Cancer Cell 2018, Table S2; doi:10.1016/j.ccell.2018.03.007.
- **Deleted arm count.** Number of chromosome arms classified as deleted; larger values indicate a greater arm-level deletion burden. *Analysed scale:* source endpoint units *Model-target source:* Taylor et al., Cancer Cell 2018, Table S2; doi:10.1016/j.ccell.2018.03.007.
- **MANTIS score.** Genome-wide microsatellite-instability score from the MANTIS algorithm; larger values indicate stronger MSI evidence. *Analysed scale:* source endpoint units *Model-target source:* Kautto et al., Oncotarget 2017; doi:10.18632/oncotarget.20432. TCGA values were obtained from cBioPortal PanCancer Atlas records.
- **MSIsensor score.** Microsatellite-instability score from MSIsensor; larger values indicate a larger fraction of unstable microsatellite loci. *Analysed scale:* source endpoint units *Model-target source:* Niu et al., Bioinformatics 2014; doi:10.1093/bioinformatics/btt755. TCGA values were obtained from cBioPortal PanCancer Atlas records.
- **Aneuploidy Score.** Aneuploidy burden distributed with the TCGA immune landscape resource; larger values indicate more arm-level copy-number alterations. *Analysed scale:* source endpoint units *Model-target source:* Thorsson et al., Immunity 2018, PanImmune_MS; doi:10.1016/j.immuni.2018.03.023.
- **Lymphocyte Infiltration Signature Score.** Gene-expression signature summarising lymphocyte-associated transcriptional infiltration; it is not a direct cell count. *Analysed scale:* source endpoint units *Model-target source:* Thorsson et al., Immunity 2018, PanImmune_MS; doi:10.1016/j.immuni.2018.03.023.
- **Nonsilent Mutation Rate.** Rate of coding mutations that alter the encoded protein. TITANPred reports the model output on the analysis log(1+x) scale. *Analysed scale:* log1p-transformed source endpoint units *Model-target source:* Thorsson et al., Immunity 2018, PanImmune_MS; doi:10.1016/j.immuni.2018.03.023.
- **SNV Neoantigens.** Predicted neoantigen burden arising from single-nucleotide variants. TITANPred reports the model output on the analysis log(1+x) scale. *Analysed scale:* log1p-transformed source endpoint units *Model-target source:* Thorsson et al., Immunity 2018, PanImmune_MS; doi:10.1016/j.immuni.2018.03.023.
- **Silent Mutation Rate.** Rate of coding mutations that do not change the encoded amino acid. TITANPred reports the model output on the analysis log(1+x) scale. *Analysed scale:* log1p-transformed source endpoint units *Model-target source:* Thorsson et al., Immunity 2018, PanImmune_MS; doi:10.1016/j.immuni.2018.03.023.
- **TIL Regional Fraction.** Estimated fraction of tumour image regions containing tumour-infiltrating lymphocytes in the TCGA immune landscape resource. *Analysed scale:* source endpoint units *Model-target source:* Thorsson et al., Immunity 2018, PanImmune_MS; doi:10.1016/j.immuni.2018.03.023.